

IB MATH AA SL

Lesson 32

The Normal Distribution

Statistics & Probability · 60-minute lesson

Syllabus SL 4.9 · Property of Lynda Badmus Education

A large, stylized, light blue letter 'N' is positioned on the right side of the slide. It is set against a dark blue background that features two large, overlapping circles of a slightly lighter shade of blue. The 'N' is composed of thick, solid strokes.

Do Now — recall

ON YOUR WHITEBOARD

1. What shape is a normal distribution?
2. Where is it centred?
3. Is it symmetric?

Answers (click to reveal)

1. a bell curve
2. at the mean μ
3. yes, about the mean

What we're learning today

LEARNING OBJECTIVES

- 1 Describe**
the normal distribution's shape.
- 2 Use**
 $X \sim N(\mu, \sigma^2)$.
- 3 Find**
probabilities with the GDC.
- 4 Apply**
the 68–95–99.7 rule.

SUCCESS CRITERIA — I CAN...

- ✓
state the properties of a normal curve
- ✓
write $X \sim N(\mu, \sigma^2)$
- ✓
find $P(X < a)$ etc. on the GDC
- ✓
use the empirical (68–95–99.7) rule

The normal curve

$$X \sim N(\mu, \sigma^2)$$

$$X \sim N(\mu, \sigma^2)$$

μ

the mean — the centre of the bell

σ

the standard deviation — the width

Area

total area under the curve = 1

68–95–99.7

68%

within 1σ of μ

95%

within 2σ

99.7%

within 3σ

Use the empirical rule

QUESTION

Heights are $N(170, 8^2)$ cm.

Between what heights do 95% of people lie?

SOLUTION

95% lie within 2σ of the mean ($\sigma = 8$):

$$170 \pm 2(8) = 170 \pm 16$$

So between:

$$154 \text{ cm} \wedge 186 \text{ cm}$$

Your turn — normal

ANSWER IN YOUR BOOK

1. $N(50, 5^2)$: 68% lie within what range?
2. Is the normal curve symmetric?
3. $P(X < \mu)$ for a normal?

Answers (click to reveal)

1. 45 to 55
2. yes
3. 0.5

GDC probabilities

FINDING AREAS

$$P(a < X < b)$$

P(X < a)

NormCdf($-\infty$, a, μ , σ)

P(X > a)

NormCdf(a, ∞ , μ , σ)

Inverse

invNorm finds the value for a given probability

TIPS

Sketch the bell and shade the region.

Symmetry

$P(X < \mu) = 0.5$.

σ vs σ^2

enter σ , not σ^2 , on the GDC.

A normal probability

QUESTION

$X \sim N(60, 5^2)$. Using a GDC, find $P(X < 65)$.

SOLUTION

65 is one σ above the mean ($60 + 5$).

NormCdf($-\infty$, 65, 60, 5):

$$P(X < 65) \approx 0.841$$

($\approx 0.5 + \text{half of } 68\% = 0.84$) ✓

Exam-style question

The mass of apples is $N(120, 10^2)$ grams.

- (a) Find $P(X < 130)$. [2]
- (b) Find the mass exceeded by 90% of apples. [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $\text{NormCdf}(-\infty, 130, 120, 10) \approx 0.841$ (M1)(A1)
- (b) $\text{invNorm}(0.10, 120, 10) \approx 107$ g (M1)(A1)(A1)

Common mistakes



σ vs σ^2

$N(\mu, \sigma^2)$ lists the variance; enter σ on the GDC.



Sketch it

Always shade the region to avoid using the wrong tail.



Symmetry

$P(X < \mu) = P(X > \mu) = 0.5$.



invNorm

Use it when given a probability and asked for a value.

Mixed practice

QUESTIONS

1. $N(100, 10^2)$: 95% range?
2. $P(X < \mu) = ?$
3. $N(20, 2^2)$: 68% range?
4. Does N list σ or σ^2 ?

Answers (click to reveal)

1. 80 to 120
2. 0.5
3. 18 to 22
4. σ^2 (variance)

Plenary & exit ticket

KEY FORMULAS

$$X \sim N(\mu, \sigma^2)$$

$$68 - 95 - 99.7$$

EXIT TICKET — before you leave

1. Centre of a normal curve?

2. $P(X > \mu)$?

3. % within 2σ ?

Answers (click to reveal)

1. the mean μ

2. 0.5

3. 95%

Self-assess:
• need more help
• nearly there
• confident

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Normal distribution.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 33

End of Topic Test — Statistics & Probability.
Revise Lessons 25–32.